Introduction

The Poisson distribution models counts of rare, independent events in a fixed window of time or space. Calls per hour, mutations per kilobase, photons per pixel, hospital admissions per day – all naturally Poisson when the underlying rate is approximately constant and events do not cluster.

Prerequisites

Discrete distributions and the notion of a rate.

Theory

A random variable X is **Poisson**(λ) if

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad k = 0, 1, 2, \dots$$

Moments:

- $E[X] = \lambda$.
- $\mathrm{Var}(X) = \lambda$ (equi-dispersion).
- Higher central moments: skewness $1/\sqrt{\lambda}$, excess kurtosis $1/\lambda$; the distribution becomes approximately normal for large λ.

Poisson limit theorem: $\mathrm{Binomial}(n, \lambda/n) \xrightarrow{d} \mathrm{Poisson}(\lambda)$ as $n \to \infty$. The Poisson is the "law of small numbers" limit of rare-event binomial trials.

Closure under sums: independent $X \sim \mathrm{Poisson}(\lambda_1)$ and $Y \sim \mathrm{Poisson}(\lambda_2)$ yield $X + Y \sim \mathrm{Poisson}(\lambda_1 + \lambda_2)$.

Connection to exponential: inter-arrival times in a Poisson process are iid exponential with rate λ.

Assumptions

The underlying events must be (approximately) independent with a constant rate. Violations of equi-dispersion (variance exceeding mean) signal **overdispersion**, commonly addressed by the negative binomial.

R Implementation

```r
library(ggplot2)

lambda <- 3.5

k <- 0:15
pmf <- dpois(k, lambda)

data.frame(k, pmf = round(pmf, 4))

# Moments
X <- rpois(1e5, lambda)
c(mean = mean(X), var = var(X), skew = mean((X - mean(X))^3) / sd(X)^3)

# Plot PMF
ggplot(data.frame(k, pmf), aes(k, pmf)) +
  geom_col(fill = "#2A9D8F") +
  scale_x_continuous(breaks = k) +
  labs(x = "k", y = "P(X = k)",
```

```r
       title = paste("Poisson(", lambda, ")")) +
  theme_minimal()

# Verify Poisson limit from binomial
n_vals <- c(10, 100, 1000, 10000)
results <- sapply(n_vals, function(n) {
  p <- lambda / n
  X <- rbinom(1e5, n, p)
  c(n = n, mean = mean(X), var = var(X))
})
t(results)
```

Output & Results

```
    k    pmf
1   0 0.0302
2   1 0.1057
3   2 0.1850
4   3 0.2158
5   4 0.1888
6   5 0.1322
...

    mean     var    skew
    3.503   3.502   0.536

         n    mean     var
[1,]    10    3.49    3.13
[2,]   100    3.50    3.40
[3,]  1000    3.50    3.49
[4,] 10000    3.50    3.50
```

Mean and variance are both ≈ 3.5 in the empirical sample; the Poisson limit from binomial tightens as $n \to \infty$.

Interpretation

Poisson regression (`glm(..., family = poisson)`) estimates log-rates from count data; reports of incidence-rate ratios and event-per-person-time summaries are typically Poisson-based. Always check for overdispersion before accepting a Poisson fit at face value.

Practical Tips

- Equi-dispersion is the signature assumption; real count data often violate it.
- Use quasi-Poisson (`family = quasipoisson`) or negative binomial (`MASS::glm.nb`) when variance exceeds mean.
- Zero-inflated Poisson (`pscl::zeroinfl`) handles excess zeros.
- For rates per unit exposure (e.g., per person-year), include an offset in the regression: `glm(y ~ x, offset = log(exposure), family = poisson)`.
- The normal approximation works for $\lambda \geq 10$; for smaller λ, use the exact PMF.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots